

CHAPTER 2

Transmission Media

Guided and unguided media · twisted pair · coaxial · optical fibre · radio, microwave, infrared, satellite · impairments and capacity · cable naming

QUICK TRICK

Learn the fast tricks first (details below):

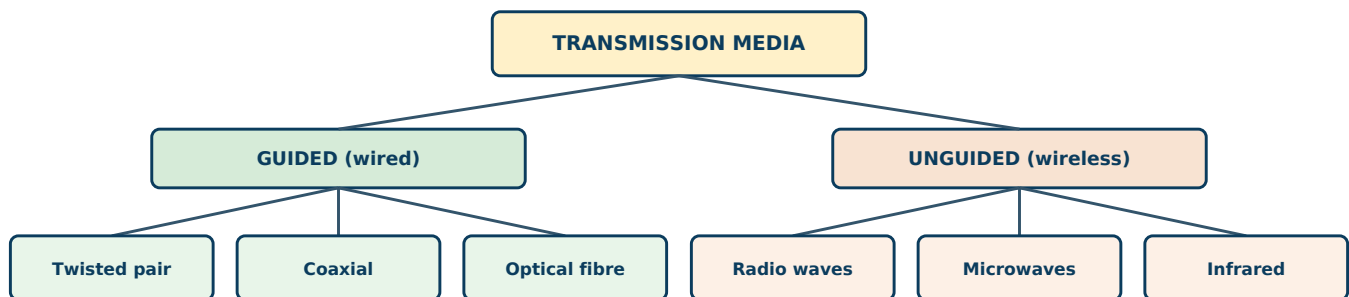
1. **Twisted pair** = copper, cheap, 100 m, RJ-45. **Coaxial** = copper core + shield, cable TV. **Fibre** = light in glass, fastest, immune to EMI.
2. **Wireless bands:** Radio 3 kHz–1 GHz (goes through walls) · Microwave 1–300 GHz (line of sight) · Infrared 300 GHz–400 THz (short range, no walls).
3. **Straight cable** joins **unlike** devices (PC-switch). **Crossover** joins **like** devices (PC-PC).
4. $\text{dB} = 10 \log_{10}(P_2/P_1)$; **Nyquist C** = $2B \log_2 L$; **Shannon C** = $B \log_2(1 + \text{SNR})$.

2.1 What is a transmission medium?

● LEVEL 1 · BEGINNER

The problem. In Chapter 1 we joined devices with a "link". Now we ask: **what is the link made of? The idea.** The **transmission medium** is the physical path between sender and receiver that carries the signal. It sits below everything else: it is the **physical layer** part of a network.

Analogy. Roads of different types. A narrow village lane (twisted pair), a highway (optical fibre) and a helicopter route through the air (wireless). All take you to the destination, but speed, cost and safety differ.



Point	Guided (wired / bounded)	Unguided (wireless / unbounded)
Path	Signal flows inside a physical conductor	Signal flows through air, vacuum or water as electromagnetic waves
Examples	Twisted pair, coaxial, optical fibre	Radio waves, microwaves, infrared, satellite
Direction	Fixed by the cable	Omnidirectional or directional by antenna
Mobility	None	Yes
Interference and security	Less interference; harder to tap (fibre best)	More interference; anyone in range can listen, so encryption is needed

QUICK TRICK

Copper cables carry **electrical** signals. Fibre carries **light** signals. Wireless carries **electromagnetic waves**.

2.2 Qualities of a medium and transmission impairments

● LEVEL 2 · EXAM

Choose a medium by looking at these factors: **bandwidth / data rate, maximum distance, cost, installation difficulty, immunity to interference and security**. Real signals also get damaged on the way. These damages are called **transmission impairments**.

Impairment	What happens	Remedy
Attenuation	Signal loses strength as it travels (more distance, more loss)	Amplifier (analog) or repeater (digital)
Distortion	Signal shape changes because different frequency parts travel at different speeds	Equalisers, better cable
Noise	Unwanted signals added. Types: thermal noise (random electron motion), crosstalk (leak from a nearby wire), impulse noise (lightning, spikes), induced noise (motors, appliances)	Twisting, shielding, good earthing
EMI	Electromagnetic interference from power lines, motors, radios	Shielded cable or fibre
Dispersion (fibre)	Light pulses spread out and overlap	Single-mode fibre, shorter distance

Decibel (dB): measuring gain and loss

dB = $10 \log_{10}(P_2 \div P_1)$ where P_1 is the input power and P_2 the output power. A negative dB means loss (attenuation), a positive dB means gain (amplification).

Power ratio P_2/P_1	0.5	0.1	2	10	100	1000
dB	-3	-10	+3	+10	+20	+30

QUICK TRICK

Half the power = **-3 dB**. Double the power = **+3 dB**. Every factor of 10 = 10 dB. Losses in dB simply **add** along a link (for example 3 dB + 5 dB = 8 dB total loss).

Signal-to-noise ratio (SNR) and channel capacity

Name	Formula	Meaning
SNR	SNR = signal power ÷ noise power; $SNR_{dB} = 10 \log_{10}(SNR)$	Higher SNR means a cleaner signal
Nyquist (noiseless channel)	$C = 2 B \log_2 L$	B = bandwidth in Hz, L = number of signal levels. Maximum bit rate of a channel with no noise.
Shannon (noisy channel)	$C = B \log_2(1 + SNR)$	Upper limit of bit rate for a channel with bandwidth B and signal-to-noise ratio SNR (a plain ratio, not dB).

Example 1. A signal of 10 mW has noise of 1 μ W. $SNR = 10 \text{ mW} \div 1 \text{ } \mu\text{W} = 10^4$, so $SNR_{dB} = 10 \log_{10}(10^4) = \mathbf{40 \text{ dB}}$.

Example 2 (Nyquist). B = 3000 Hz and 4 signal levels. $C = 2 \times 3000 \times \log_2 4 = 2 \times 3000 \times 2 = \mathbf{12 \text{ 000 bps}}$.

Example 3 (Shannon). B = 3000 Hz and SNR = 1023. $C = 3000 \times \log_2(1 + 1023) = 3000 \times 10 = \mathbf{30 \text{ 000 bps}}$.

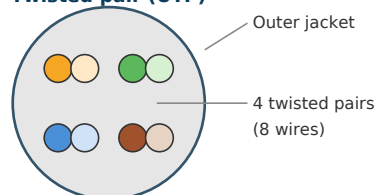
COMMON MISTAKE

- Putting SNR in dB directly into Shannon's formula. Convert to a plain ratio first: $SNR = 10^{(dB/10)}$. 30 dB means a ratio of 1000, not 30.
- Using Nyquist for a noisy channel. Nyquist ignores noise; Shannon includes it. The real limit is the smaller of the two.

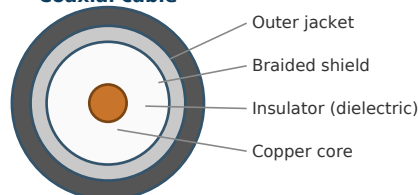
2.3 Guided media (1): twisted pair cable

● LEVEL 1 • BEGINNER

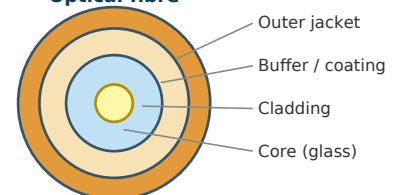
Twisted pair (UTP)



Coaxial cable



Optical fibre



Twisted pair has pairs of insulated copper wires twisted around each other. The twisting makes the noise that hits both wires equal, so it cancels at the receiver. It also reduces **crosstalk** between neighbouring pairs. More twists per inch means better quality.

Point	UTP (Unshielded Twisted Pair)	STP (Shielded Twisted Pair)
Structure	Pairs inside a plastic jacket, no metal shield	Each pair (or the whole bundle) covered with metal foil or braid
Interference	More affected by EMI	Better protected
Cost and weight	Cheap, light, thin, easy to install	Costlier, heavier, thicker, needs proper earthing
Use	Almost all office and home LANs, telephone	Factories, places with heavy electrical noise
Connector	RJ-45 (8 pins) for LAN; RJ-11 (telephone)	RJ-45 shielded

Category	Typical speed	Bandwidth (MHz)	Note
Cat 3	10 Mbps	16	Old telephone and 10BASE-T
Cat 5	100 Mbps	100	100BASE-TX Fast Ethernet
Cat 5e	1 Gbps	100	1000BASE-T; very common
Cat 6	1 Gbps (10 Gbps up to about 55 m)	250	Better crosstalk control
Cat 6a	10 Gbps up to 100 m	500	Offices, data centres
Cat 7	10 Gbps	600	Fully shielded; not common in homes
Cat 8	25 / 40 Gbps up to about 30 m	2000	Data-centre links

BPSC POINT

Maximum length of one UTP segment in Ethernet = 100 m. A cable has **4 pairs = 8 wires**; 10/100 Mbps Ethernet uses only 2 pairs, gigabit Ethernet uses all 4.
Higher category number = higher speed and bandwidth.

Straight-through and crossover cables, T568A and T568B

Both ends of an RJ-45 cable are wired by a colour code. There are two standards, **T568A** and **T568B**. T568B is the one most often used in India and the US for LAN cables.

Pin	T568A colour	T568B colour	Used for 10/100 Mbps
1	white-green	white-orange	TX+
2	green	orange	TX–
3	white-orange	white-green	RX+
4	blue	blue	unused (10/100)
5	white-blue	white-blue	unused (10/100)
6	orange	green	RX–
7	white-brown	white-brown	unused (10/100)
8	brown	brown	unused (10/100)

Cable	Wiring	Connects	Example
Straight-through	Same standard on both ends (B-B or A-A)	Unlike devices	PC ↔ switch, PC ↔ router, switch ↔ router
Crossover	T568A on one end, T568B on the other	Like devices	PC ↔ PC, switch ↔ switch, router ↔ router

QUICK TRICK

Rule: **like devices → crossover; unlike devices → straight**. Modern network cards and switches have **auto MDI-X** and sort it out themselves, so a straight cable works almost everywhere today.

Advantages of twisted pair: cheap, flexible, easy to install and repair, widely supported. **Disadvantages:** limited distance (100 m), lower bandwidth than fibre, UTP is sensitive to EMI and can be tapped.

2.4 Guided media (2): coaxial cable

● LEVEL 1 • BEGINNER

Coaxial cable ("coax") has a solid copper **core** in the centre, surrounded by an **insulator** (dielectric), then a metal **braided shield**, and finally a plastic **outer jacket**. The shield and the core share the same axis, which is where the name "co-axial" comes from. The shield protects the signal from outside noise and stops the signal leaking out.

Type	Standard / use	Impedance	Maximum segment	Connector
Thinnet (RG-58)	10BASE2, old LANs	50 Ω	185 m (about 200 m), up to 30 nodes	BNC, T-connector, terminator
Thicknet (RG-8)	10BASE5, old LAN backbone	50 Ω	500 m, up to 100 nodes	N-type, vampire tap
RG-59 (and RG-6)	Cable TV, CCTV, broadband cable	75 Ω	Depends on use	F-type, BNC

Baseband coax	Broadband coax
Digital signal, one channel at a time on the whole cable, 50 Ω (Ethernet bus)	Analog signal, many channels by frequency division, 75 Ω (cable TV)

Advantages: better shielding and higher bandwidth than twisted pair, good for longer runs, durable.

Disadvantages: bulkier and costlier than UTP, harder to install, a break in a bus stops the network, almost no longer used in new LANs.

QUICK TRICK

Coax = cable TV and old bus Ethernet. 50 Ω = data (10BASE2, 10BASE5). 75 Ω = TV and video.

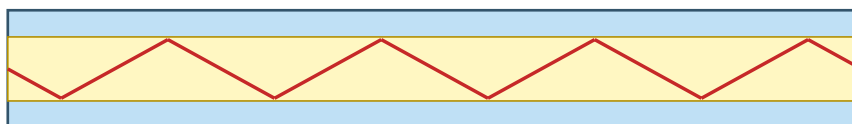
2.5 Guided media (3): optical fibre cable

● LEVEL 1 • BEGINNER

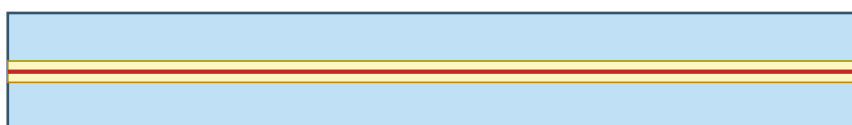
Optical fibre sends data as pulses of **light** through a very thin strand of glass or plastic. It has three main parts: the **core** (where light travels), the **cladding** (glass around the core with a lower refractive index) and a protective **buffer coating and jacket**. A strand is about as thin as a human hair (cladding 125 μm).

Light stays inside the core by total internal reflection

Multimode
(many paths)



Single-mode
(one path)



Principle: total internal reflection (TIR). When light goes from a denser medium (core, index n_1) to a rarer one (cladding, $n_2 < n_1$) at an angle bigger than the **critical angle**, it does not leave: it reflects back into the core. **$\sin \theta_c = n_2 \div n_1$** . The ray bounces along the core to the other end.

Example. Core $n_1 = 1.5$, cladding $n_2 = 1.0$. $\sin \theta_c = 1.0 \div 1.5 = 0.667$, so $\theta_c \approx 41.8^\circ$. Light speed in this glass = $c \div n = 3 \times 10^8 \div 1.5 = 2 \times 10^8 \text{ m/s}$.

Point	Single-mode fibre (SMF)	Multimode fibre (MMF)
Core diameter	About 8–10 μm (thin)	50 or 62.5 μm (thick)
Light path	Only one ray (mode), straight	Many rays (modes) bouncing at different angles
Light source	Laser diode	LED (or VCSEL laser)
Wavelength	1310 nm and 1550 nm	850 nm and 1300 nm
Distance	Very long: tens to over 100 km	Short: hundreds of metres to about 2 km
Dispersion	Very low	Higher (pulses spread)
Cost	Higher (laser, precise core)	Lower
Use	Telecom backbone, undersea cables, ISP links	Inside buildings, campus, data centres

Multimode fibre is of two kinds: **step-index** (sharp change at the core edge, rays zigzag) and **graded-index** (index falls smoothly from centre to edge, rays curve and arrive almost together, so less dispersion).

Advantages	Disadvantages
Huge bandwidth (Tbps possible) Very low attenuation (about 0.2 dB/km at 1550 nm) Immune to EMI and crosstalk Very secure, hard to tap Thin, light, no spark risk	Costly cable and equipment Fragile; bends sharply cause loss Joining (splicing) and installation need skill One-way light, so two fibres are needed for two-way Needs electrical $\leftrightarrow$ optical conversion

Common connectors: SC (square snap-in), ST (round twist-lock), LC (small form factor) and FC. **Light source** at sender: LED or laser. **Detector** at receiver: photodiode.

Example (fibre loss). A 50 km link with loss 0.2 dB/km has $50 \times 0.2 = 10$ dB total loss. Power drops by a factor of 10, so 1 mW at the input becomes **0.1 mW** at the output.

QUICK TRICK

Remember "**single = thin, laser, long**" and "**multi = thick, LED, short**".

2.6 Comparison of guided media

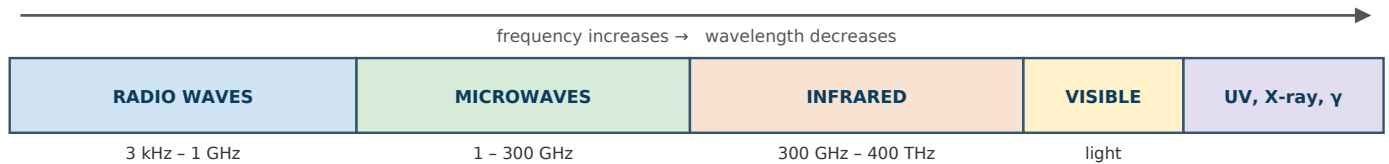
● LEVEL 2 • EXAM

Point	Twisted pair (UTP)	Coaxial	Optical fibre
Signal	Electrical	Electrical	Light
Typical data rate	10 Mbps – 10 Gbps	Up to about 10 Mbps (LAN); more in cable broadband	Gbps – Tbps
Distance	100 m per segment	185–500 m per segment (LAN)	Kilometres, no repeater needed for tens of km
Attenuation	High	Medium	Very low
EMI immunity	Low (UTP)	Medium	Complete
Cost	Lowest	Medium	Highest
Installation	Easy	Moderate	Difficult
Security from tapping	Low	Medium	High
Main use	LAN, telephone	Cable TV, old LAN	Backbone, WAN, long links

2.7 Unguided media: wireless transmission

● LEVEL 1 • BEGINNER

In wireless transmission, signals travel through free space as **electromagnetic waves**. No cable is needed, so devices can move. The waves are produced and received by **antennas**. The usable range of frequencies is called the **electromagnetic spectrum**. The three bands used for communication are:



Band	Frequency	Behaviour	Examples of use
Radio waves	3 kHz – 1 GHz	Omnidirectional (spread in all directions); can pass through walls; travel long distances	AM/FM radio, TV broadcast, cordless phones, paging
Microwaves	1 GHz – 300 GHz	Unidirectional, narrow beam; need line of sight; cannot pass walls; rain can weaken them	Mobile networks, Wi-Fi, satellite links, radar, microwave towers
Infrared	300 GHz – 400 THz	Short range, line of sight; cannot pass walls; no licence needed; sunlight disturbs it	TV remote, IrDA, short-range data

Frequency bands (finer divisions)

Band	Range	Typical use
VLF	3 – 30 kHz	Submarine communication, navigation
LF	30 – 300 kHz	Navigation, time signals
MF	300 kHz – 3 MHz	AM radio (530–1700 kHz)
HF	3 – 30 MHz	Shortwave radio, amateur radio
VHF	30 – 300 MHz	FM radio (88–108 MHz), TV
UHF	300 MHz – 3 GHz	TV, mobile phones, Wi-Fi 2.4 GHz, Bluetooth
SHF	3 – 30 GHz	Satellite, radar, Wi-Fi 5 GHz
EHF	30 – 300 GHz	Short links, 5G millimetre-wave, radio astronomy

How waves travel (propagation)

Mode	How	Used by
Ground wave	Follows the curve of the Earth near the surface	Very low and low frequencies (up to about 2 MHz)
Sky wave	Reflects from the ionosphere and returns to Earth	Shortwave, 2–30 MHz; long distance across continents
Line of sight	Straight path between two antennas that can "see" each other	VHF and above: microwave, satellite, mobile, Wi-Fi

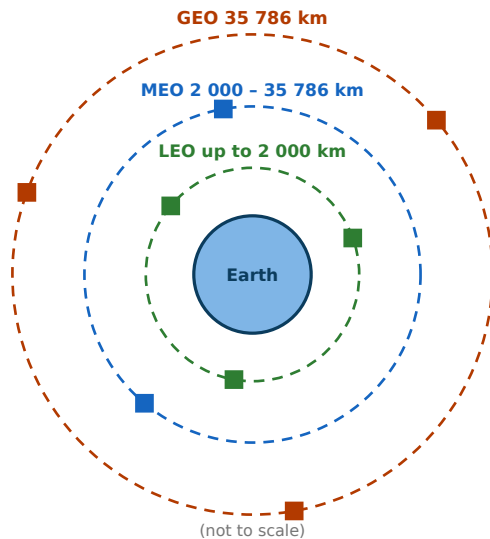
BPSC POINT

Microwave towers are placed every **about 40–70 km** with antennas on tall towers, because the Earth curves and the beam is line of sight. Repeaters on the towers receive and re-send the signal.

Radio = omnidirectional, passes walls. Microwave = unidirectional, no walls. Infrared = very short range, no walls.

Satellite communication

A communication satellite is a microwave repeater in space. The Earth station sends the signal **up** (uplink); the **transponder** on the satellite amplifies it and changes its frequency; it is then sent **down** (downlink) to another Earth station. Different frequencies for uplink and downlink avoid interference.



GEO (Geostationary). About 35 786 km above the equator. One orbit takes 24 h, so the satellite looks fixed in the sky and a dish never needs to move. Three satellites can cover the Earth. Used for DTH TV, weather and INSAT/GSAT-type services. Delay is large.

MEO (Medium Earth Orbit). About 2 000 to 35 786 km. GPS satellites are at about 20 200 km. Orbit takes several hours.

LEO (Low Earth Orbit). Up to about 2 000 km (many broadband constellations fly near 500–1 200 km). Low delay and low power, but each satellite sees a small area, so many satellites are needed.

Example (GEO delay). One-way, Earth to satellite = $35\,786\text{ km} \div 3 \times 10^5\text{ km/s} \approx 0.119\text{ s}$. A hop Earth → satellite → Earth $\approx 2 \times 0.119 \approx \mathbf{0.24\text{ s}}$ (about 240 ms). A question and its reply take roughly 0.5 s, which is why GEO internet feels slow in calls.

Short-range wireless technologies

Technology	Band / frequency	Range	Standard / note
Bluetooth	2.4 GHz	About 10 m (class 2)	IEEE 802.15.1, wireless PAN
Wi-Fi	2.4, 5 and 6 GHz	Tens of metres indoors	IEEE 802.11, wireless LAN
NFC	13.56 MHz	A few cm (about 4 cm)	Contactless payment, tags
Infrared (IrDA)	Infrared band	Within a metre or a few metres, line of sight	Remote controls
Zigbee	2.4 GHz (also lower bands)	10–100 m	IEEE 802.15.4, low-power IoT

Cellular generations (1G to 5G) are covered later in the chapter on mobile communication.

2.8 Wired vs wireless, and choosing a medium

● LEVEL 2 • EXAM

Point	Wired	Wireless
Speed and stability	High and steady	Variable; affected by distance, walls, weather
Security	Better (need physical access)	Easier to intercept; needs encryption
Mobility	None	Excellent
Installation cost	Cable laying is costly	Cheap to set up in difficult places
Interference	Low	High (other devices, same band)
Expansion	Needs new cable	Easy

Need	Best choice	Reason
Computer lab LAN	UTP (Cat 5e / 6)	Cheap, easy, 100 m is enough
Link between two buildings 5 km apart	Single-mode fibre	Long distance, immune to EMI and lightning surges
Cable TV connection	Coaxial	75 Ω, good shielding
Factory with heavy motors	STP or fibre	High electrical noise
Remote hill village, no cables	Microwave or satellite	No ground cable needed
TV remote, phone to earbuds	Infrared / Bluetooth	Very short range
Backbone between cities	Fibre	Huge bandwidth, low loss

2.9 Ethernet cable naming (decode the name)

● LEVEL 2 · EXAM

Names like **10BASE-T** have three parts: **speed in Mbps** + **signalling** (BASE = baseband, one digital channel) + **medium** (a number means about how many hundreds of metres, a letter means the medium).

Name	Speed	Medium	Max segment
10BASE5	10 Mbps	Thick coax (thicknet)	500 m
10BASE2	10 Mbps	Thin coax (thinnet)	185 m (about 200 m)
10BASE-T	10 Mbps	Twisted pair (Cat 3 or better)	100 m
100BASE-TX	100 Mbps (Fast Ethernet)	Twisted pair, Cat 5	100 m
100BASE-FX	100 Mbps	Multimode fibre	About 2 km
1000BASE-T	1 Gbps	Twisted pair, Cat 5e or better	100 m
1000BASE-SX / LX	1 Gbps	Fibre (SX short wave, multimode; LX long wave)	Hundreds of metres to kilometres
10GBASE-T	10 Gbps	Twisted pair, Cat 6a	100 m

QUICK TRICK

T = twisted pair, F = fibre, 2 and 5 = coax (200 m and 500 m). The first number is always the speed in Mbps.

2.10 Common mistakes

● LEVEL 2 · EXAM

COMMON MISTAKE

- Saying fibre carries electricity. It carries **light**.
- Mixing the cable rule: **like devices need crossover**, unlike devices need straight (not the other way round).
- Writing that radio waves cannot pass walls. Radio passes through walls; **microwave and infrared do not**.
- Using SNR in dB inside Shannon's formula. Convert to a ratio first.
- Forgetting that the 100 m limit is for a **twisted pair segment**, not for fibre.
- Thinking multimode means "better". Multimode is shorter range and cheaper; single-mode goes much farther.
- Confusing GEO (high, fixed, big delay) with LEO (low, moving, small delay).

Quick-Solve Tricks — Summary

Situation	Trick
Guided vs unguided	Guided = cable (twisted pair, coax, fibre). Unguided = air (radio, microwave, infrared).
Twisted pair	100 m, RJ-45, Cat 5e = 1 Gbps, twist cancels noise
Straight / crossover	Unlike devices = straight; like devices = crossover
Coax	50 Ω data (10BASE2, 10BASE5); 75 Ω TV; BNC and T-connector
Fibre	Light, TIR, core + cladding; single-mode thin laser long; multimode thick LED short
Critical angle	$\sin \theta_c = n_2 \div n_1$
Wireless bands	Radio 3 kHz–1 GHz · Microwave 1–300 GHz · Infrared 300 GHz–400 THz
Walls	Radio passes; microwave and infrared do not
Satellite	GEO 35 786 km (24 h, fixed) · MEO 2 000–35 786 km · LEO below 2 000 km
dB	$10 \log_{10}(P_2/P_1)$: half = -3 dB, $\times 10$ = $+10$ dB
Capacity	Nyquist $2B \log_2 L$ (no noise) · Shannon $B \log_2(1 + \text{SNR})$ (noise)
Naming	10BASE5 = 10 Mbps, baseband, 500 m · 10BASE-T = twisted pair

BPSC QUICK CHECK — Chapter 2

UTP max length → 100 m	RJ-45 → LAN, RJ-11 → telephone
Coax core + dielectric + shield + jacket	Fibre core + cladding + coating + jacket
Fibre is immune to EMI	Fibre uses total internal reflection
Single-mode → 9 μm core, laser	Multimode → 50/62.5 μm core, LED
Radio → through walls	Microwave → line of sight, towers
GEO → 35 786 km, appears fixed	Bluetooth → 2.4 GHz, 802.15.1

Traps: straight vs crossover · SNR in dB vs ratio · radio vs microwave through walls · GEO vs LEO delay · single-mode vs multimode · 50 Ω vs 75 Ω coax.

Practice questions

● LEVEL 1 · BEGINNER

- **Q1.** What is a transmission medium? Name its two categories.
- **Q2.** Which cable has a copper core, insulator, braided shield and outer jacket?
- **Q3.** Which wired medium is not affected by electromagnetic interference?
- **Q4.** Which connector is used with UTP cable in a LAN?
- **Q5.** Name three unguided media.
- **Q6.** Why are the two wires of a twisted pair twisted?

● LEVEL 2 · EXAM

- **Q7.** Which cable do you use to join two computers directly (classic Ethernet), and why?
- **Q8.** Give three differences between single-mode and multimode fibre.
- **Q9.** What does 10BASE5 mean?
- **Q10.** A fibre has core index 1.5 and cladding index 1.0. Find the critical angle.
- **Q11.** Why are microwave towers built at fixed distances and on tall towers?
- **Q12.** Why does a GEO satellite appear fixed in the sky? Which orbit, GEO or LEO, gives lower delay?

● LEVEL 3 · BPSC / PGT

- **Q13.** Signal power falls from 20 mW to 2 mW along a cable. Find the loss in dB.
- **Q14.** A noiseless channel has $B = 4000$ Hz and 8 signal levels. Find the maximum bit rate (Nyquist).
- **Q15.** A channel has $B = 1$ MHz and $\text{SNR} = 255$. Find the capacity (Shannon).
- **Q16.** A fibre link is 80 km long and loses 0.25 dB/km. If 10 mW enters, how much power comes out?
- **Q17.** Find the one-way delay from Earth to a GEO satellite (signal speed 3×10^8 m/s) and the Earth-satellite-Earth delay.
- **Q18.** An SNR is given as 30 dB. What is the plain ratio, and what is the Shannon capacity for $B = 3000$ Hz? (take $\log_2 1001 \approx 10$)

Answers

Q	Solution
1	The physical path that carries the signal from sender to receiver. Categories: guided (wired) and unguided (wireless).
2	Coaxial cable.
3	Optical fibre.
4	RJ-45.
5	Radio waves, microwaves, infrared (satellite also uses microwaves).
6	Noise hits both wires almost equally and cancels at the receiver; twisting also reduces crosstalk between pairs.
7	A crossover cable (T568A at one end, T568B at the other), because the transmit pins of one PC must meet the receive pins of the other. Modern cards with auto MDI-X work with a straight cable too.
8	Single-mode: $\sim 9\text{ }\mu\text{m}$ core, laser, one ray, very long distance. Multimode: 50/62.5 μm core, LED, many rays, short distance, more dispersion.
9	10 Mbps, baseband signalling, maximum segment about 500 m (thick coax, thicknet).
10	$\sin \theta_c = 1.0 \div 1.5 = 0.667$, so $\theta_c \approx 41.8^\circ$.
11	Microwaves travel in straight lines (line of sight) and the Earth curves, so towers must be tall and spaced about 40–70 km, with repeaters re-sending the signal.
12	Its orbit time is 24 h, equal to the Earth's rotation, and it is over the equator, so it stays above one point. LEO gives lower delay because it is much closer.
13	$\text{dB} = 10 \log_{10}(2 \div 20) = 10 \log_{10}(0.1) = -10\text{ dB}$ (a loss of 10 dB).
14	$C = 2 \times 4000 \times \log_2 8 = 2 \times 4000 \times 3 = 24\text{ }000\text{ bps}$.
15	$C = 10^6 \times \log_2(1 + 255) = 10^6 \times \log_2 256 = 10^6 \times 8 = 8\text{ Mbps}$.
16	Loss = $80 \times 0.25 = 20\text{ dB}$, a factor of 100. Output = $10\text{ mW} \div 100 = 0.1\text{ mW}$.
17	One-way = $35\text{ }786\text{ }000\text{ m} \div 3 \times 10^8 \approx 0.119\text{ s}$ (about 119 ms). Earth $\rightarrow$ satellite $\rightarrow$ Earth $\approx 2 \times 0.119 \approx 0.24\text{ s}$.
18	Ratio = $10^{(30/10)} = 1000$. $C = 3000 \times \log_2(1 + 1000) \approx 3000 \times 10 = 30\text{ }000\text{ bps}$.

REMEMBER

Next: Chapter 3 — Network Devices (modem, NIC and MAC address, repeater, hub, bridge, switch, router, gateway, access point, and which OSI layer each works at).

Roadmap: Ch 4 Switching, Multiplexing and Error Detection · Ch 5 OSI and TCP/IP Models · Ch 6 Network Protocols · Ch 7 IP Addressing and Subnetting · Ch 8 Web Services · Ch 9 Network Security · Ch 10 Mobile Telecom, Cloud and IoT.